DEPARTMENT OF MATHEMATICS & COMPUTING

End Semester Examination

Session: Winter Semester, 2024-2025
Subject: Engineering Mathematics-II (NMCI102)

Duration: 2 Hours
Marks: 50

Instructions:

- All questions are compulsory. **Calculators** are not allowed.
- Figures in the margin indicate marks for the concerned questions.
- Notations have their usual meaning.

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1. Find the linear transformation $T\colon \mathbb{R}^3 \to \mathbb{R}^2$ defined by $T(1,0,0) = (2,1), T(0,1,0) = (0,1)$, and $T(0,0,1) = (1,1)$. Find the range space, kernel, rank, and nullity of T. [7]

2. (a) For the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$, find the value of $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$, where I is a 3×3 identity matrix. [3]

   (b) Let $B = \begin{bmatrix} 2 & -2 & 0 \\ -2 & 1 & -2 \\ 0 & -2 & 0 \end{bmatrix}$. Find an orthogonal matrix P such that $P^{-1}BP$ is a diagonal matrix. [4]

3. Let $y_1(x) = 1$, $y_2(x) = \dfrac{x^3}{3}$ and $y_3(x)$ be three linearly independent solutions of the differential equation:

$$P_3(x)\frac{d^3y}{dx^3} + P_2(x)\frac{d^2y}{dx^2} + P_1(x)\frac{dy}{dx} + P_0(x)y = 0, \quad 0 < x < 2,$$

   where $P_0(x), P_1(x), P_2(x)$, and $P_3(x)$ are continuous functions on the interval $(0,2)$. If $y_3(1) = 0$, $\dfrac{dy_3}{dx}(1) = -\frac{1}{5}$ and the Wronskian $W(y_1, y_2, y_3) = \dfrac{\cos(2\log_e x)}{x}$, $0 < x < 2$, then find $y_3(x)$. [7]

4. Solve the following initial value problem: [7]

$$xy' + 2y = -x^3y^2\cos x, \quad y(1) = 2.$$

$x\frac{dy}{dx} + 2y = -x^3y^2\cos x$

$x\,dy + (2y + x^3y^2\cos x)\,dx = 0$

5. Consider the initial value problem:

$$\frac{dy}{dx} = \frac{y}{1+x^2}, \quad y(0) = 1.$$

   (a) Show that $f(x,y) = \dfrac{y}{1+x^2}$ satisfies the Lipschitz condition in y on the rectangle

   $R = \{(x,y) \in \mathbb{R}^2 : |x| \le 1,\ |y| \le 2\}$ [1]

   (b) Use Picard's theorem to justify whether a unique solution exists near $x = 0$. If yes, then find an interval around $x = 0$ where the solution is guaranteed to exist. [3]

   (c) Starting with $y_0(x) = 1$, compute the first two iterations of Picard's method. [3]

$\ln y = \tan^{-1} x + 1$

$(\lambda - 4)(\lambda + 2)$

[P. T. O.]

6. (a) Find the general solution of the differential equation, $x^3y'' - 2xy = 6\ln x;\ x > 0$. [4]

(b) Find the general solution of the following equation by the method of variation of parameters. [3]

$$y'' - 5y' + 6y = 2e^x.$$

7. (a) Form a partial differential equation from $z(x, y) = y^2 + 2f(\frac{1}{x} + \ln y)$, where f is an arbitrary function. $x > 0$. and $y > 0$. [3]

(b) Solve the partial differential equation $x^2\dfrac{\partial z}{\partial x} + y^2\dfrac{\partial z}{\partial y} = (x + y)z$. [5]